

CS-351 PROJECT (2022-2023)

Room Reservation Management System

Brief description of the project

In this project, you are asked to analyze and design a simplified room reservation management system. In summary, the system users are the employees of a company, who submit requests to reserve rooms for meetings or events, and the administrators, who are asked, among other things, to approve those reservation requests.

The aim of this project is to familiarize yourselves with the basic concepts and principles of information systems (IS) analysis and design. The project will be carried out by groups of two people, while each member will be evaluated separately, depending on their participation, after an oral examination.

This work is divided into two phases (P1 and P2), while parts of P1 have been partially covered in the exercises (A1, A2), as described below:

PHASE 1 (P1): Analysis

In short, this phase should include the following:

- Requirements Elicitation (Extension of A1 (a))
- Data Modeling – ER Diagram (Extension of A1 (c))
- Class Diagram (Extension of A2 (b))

PHASE 2 (P2): Design and Rudimentary Implementation

In short, Phase 2 should include the following:

- Database Design (logical level)
- Interface and Dialogue Design
- Rudimentary Implementation
- Documentation

15% Bonus for P2: Full Implementation with GUI.

Step-by-step instructions follow next.

PHASE 1 (P1): Analysis

The aim of Phase 1 is to analyse a simplified room reservation management system. Specifically, you are asked to expand parts of exercises A1 and A2 to support all the functions that your system will provide.

Steps. The steps you will need to follow are as follows:

1. Requirements Elicitation. Collect and organize functional and non-functional requirements (numbering, short description, type of non-functional requirements) [20%]
2. Data Modeling: ER Diagram (ERD) [40%]
3. Detailed Class Diagram [40%]

Note that the steps are not independent of each other, but parts of the same system. Therefore, your responses to the individual steps (and to Phase 2 later) should be consistent with each other.

Step 1: Set all the functional requirements that your system will provide (maximum ten), including (optionally) what you set in Assignment 1. For each functional requirement, you will need to define 1-3 corresponding non-functional requirements, as well as the type of each non-functional requirement.

Step 2: Expand the ERD of Assignment 1 to include all the needed entities, their attributes, and all the relationships between them. You should also include the cardinality ratios of the relationships.

Step 3: Expand (if necessary) the class diagram you submitted in Question (b) of Exercise 2 to cover everything you defined above. You should include the necessary operations, attributes and relationships between the classes.

Deliverables and submission instructions: The deliverable of this phase is a **pdf report**, which will clearly show how you handled each of the above steps. For steps that require diagrams, you should include (in the same pdf file) generated images from the Modelio tool, or from another CASE tool of your choice.

Name the file you will submit to e-learn as follows: **TeamXX_StudentId1_LastName1_StudentId2_LastName2.pdf**, where XX is your team ID in e-learn, StudentId 1 and LastName1 are the student id and surname, respectively, of the member who submitted the file, on behalf of the whole team.

PHASE 2 (P2): Design and Rudimentary Implementation

The purpose of Phase 2 is to familiarize yourselves with the process of designing an information system. Specifically, you are asked to design the information system you dealt with in the previous phase and implement a small part of it.

Requested

The requirements of P2 are the following:

1. Database Design (logical level) [35%]
2. Interface and Dialogue Design [20%]
3. Rudimentary Implementation [40%] + Full Implementation Bonus [15%]
4. Documentation [5%]

Note that those four requirements are not independent of each other, but are part of the same system. Therefore, your answers to the individual questions should be consistent with each other, as well as with your answers for P1. If you see fit to make changes to your P1 responses, you should include them and clarify what those changes are and why they were made.

Requirement 1: Use the techniques you were taught in the course to convert Entity-Relationship Diagrams (ERDs) into normalized relations. In your answer, you should include the ERD of P1, highlighting any changes that may have been made, as well as the normalized relations that result from them, with the primary and foreign keys evident.

Requirement 2: Draw your system's interface and dialogue diagram.

Step 1 [10%]. You can design the screens using the Figma tool (<https://www.figma.com/>), or Powerpoint.

Step 2 [10%]. Draw the dialogue diagram for the screens you drew in Step 1.

Requirement 3: Provide a rudimentary implementation for your system.

Step 1 [3%]. Modify the class diagram of P1, if necessary, and include it, with any changes marked clearly, in your answer.

Step 2 [2%]. Provide the package diagram of your system classes.

Step 3 [35%]. Rudimentary implementation of (parts of) your system in Java. Connect your program to data (either simple data in one or more text files, or data from a Database), according to Requirement 1. Specifically, you are asked to implement the use cases:

"Room reservation request"

"View room availability"

"Approve reservation request (by administrator)"

(as in A2)

as well as two more use cases.

Attention: The two additional use cases should not be linked either to each other or to any other use case than the above through "includes" or "extends".

Step 4 **[15% Bonus]**. Full implementation

Step 4.1 **[5% Bonus]** The implementation of an operational graphical interface (as in Requirement 2), is considered a **Bonus (5%)**.

Step 4.2 **[10% Bonus]** If you implement a complete system (according to all your functional and non-functional requirements, as defined in Step 1 of Phase 1, you will receive a **10% Bonus**.

Requirement 4 [5%]: Clear instructions for installing and executing the code (what operating system it is running on, which Java version, what external libraries are needed, etc.). Internal documentation (within the code, not in a separate file) is also expected, i.e., clear comments describing what your code does.

Deliverable and submission instructions:

A zip file containing an **"implementation" folder** in which you should include the code you wrote for Requirement 3 (which should also include internal documentation for Requirement 4). In the same zip file, you should also include a **pdf report** that covers Requirements 1, 2, 4 (without internal documentation).

Name the zip file you will submit to e-learn as follows: **TeamXX_P2_StudentId1_LastName1_StudentId2_LastName2.zip**, where XX is your team ID in e-learn, StudentId 1 and LastName1 are the student id and surname, respectively, of the member who submitted the file, on behalf of the whole team.